Experiment :- Study of LAN Topology, Network devices and Address configuration.

Objective :-

1.) To study the LAN topology used in college campus.

2.) To identify networking device & understand their function.

3.) To analyze wired & wireless communication method used in the network

4.) To measure speed of network for wired & wireless connections.

5.) To obtained MAC, IP Address & subnet mask.

Problem Statement - 1

- Draw & identify LAN Topology of college campus and clearly label all components such as router, switches & etc.

Soln :-

The College Campus network observed this study consists of :

* Layer - 3 Core switches in the Server Room.
* Managed by POE + Switches.
* Wireless Access Point (APs)
* Ethernet port in classroom & laboratories.

Network Topology

Hybrid Hierarchical Topology which combines:

1.) Tree Topology
2.) Extended Star Topology
3.) Ring Topology (Server Room).

The network hierarchical layer

(1) Core Layer
(2) Distribution Layer
(3) Access Layer

Core Layer

* Core layer contains server Room and Server room and Server room have Firewall, Router, 2-L3+ Switches, Patch Rack, Server, inverter etc.

Distribution Layer

* Distribution layer contains 4-switches.
* The 4 switches are →
  (1) Main Build → R7 (Room no.-213, 2nd floor)
  (2) Main Build → R1 (Room no.-301, 3rd floor)
  (3) Main Build → R6 (Room no.-113, 1st floor)
  (4) Main Build → R2 (Room no.-201, 2nd floor)
* This 4-switches contains →
  (1) 18 port Gigabit Easy Smart Switch with 16-port. TL-SG1218MPE

② SG 3428 X MP → 24 Port Gigabit & 4 port ~~10~~ 10 GE SFP+, L2+ Managed switches with 24 Port POE+.

→ Now let's discuss these 2 above components

① TP-Link TL-SG 1218MPE

- likely used for: CCTV Camera, Wifi access point, etc.
- 18 Port Gigabit "Easy Smart" Switch,
- 16 port Support POE +

② TP-Link Omada SG3428 X MP

- Higher-end managed layer 2+ Switch,
- 24 POE+ Port, Uplinks, 10 G SFP
- Usually acts as a main distribution for each floor.

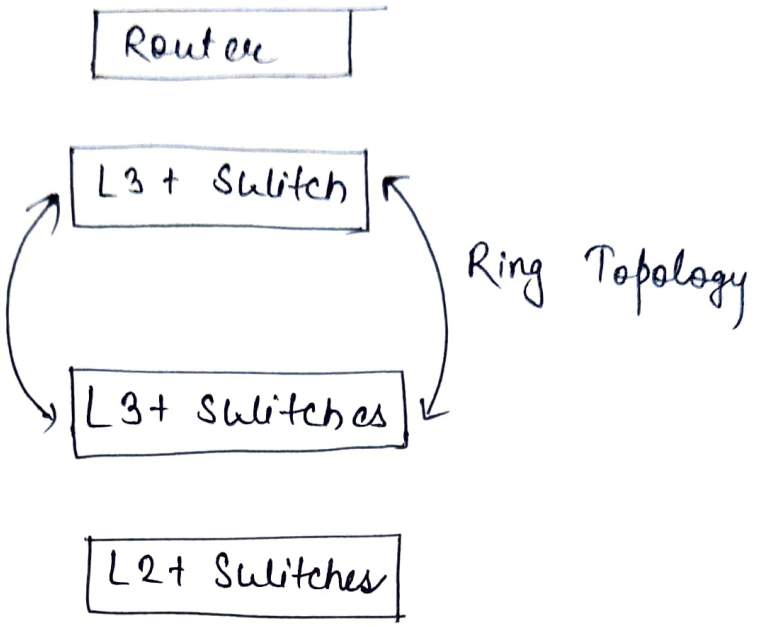
→ Access layer:

- Access layer contain 3-switches i.e

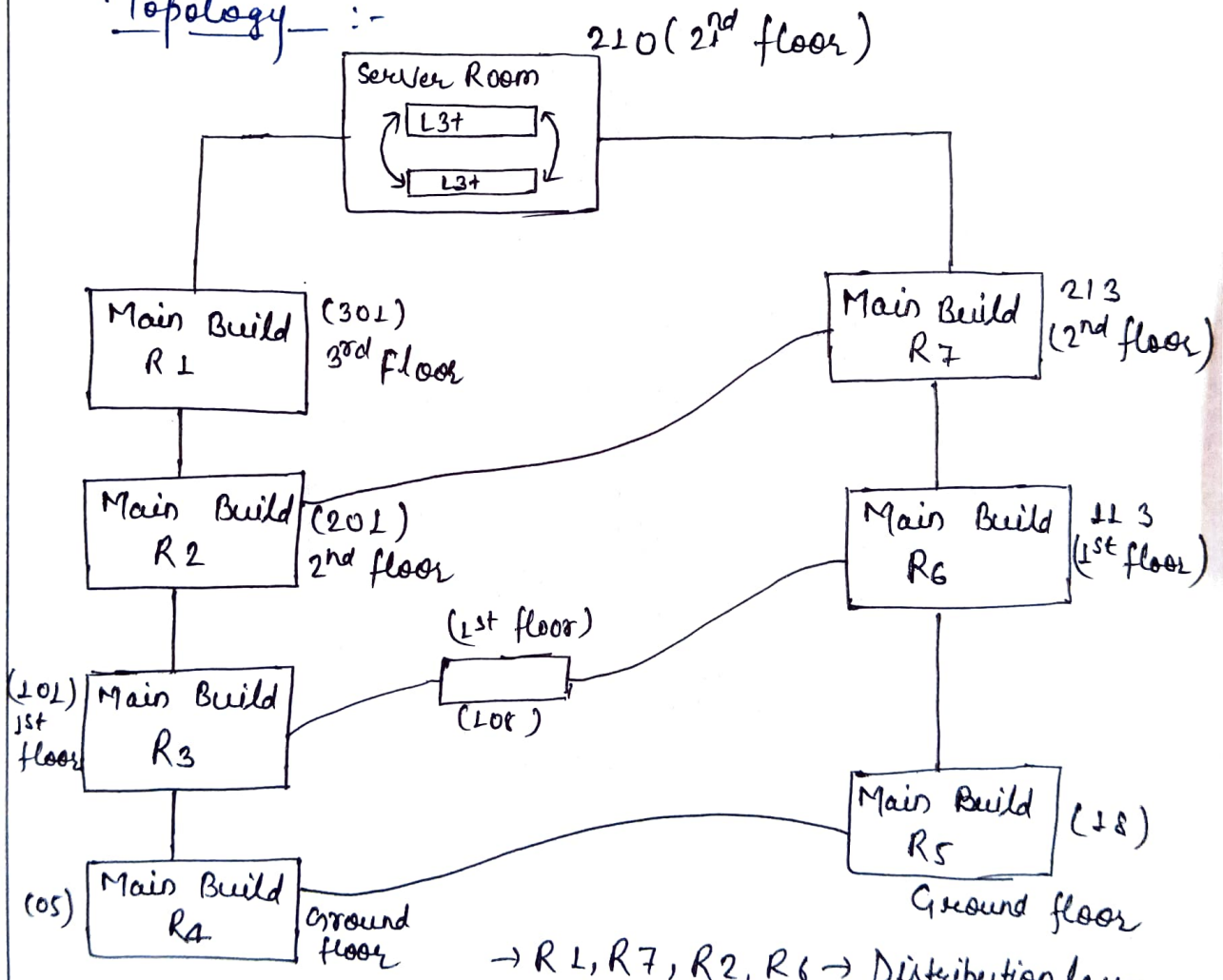
1.) Main Build R1 - [Room no.- 301, 3rd Floor]
2.) Main Build R4 - [Room no.- 05, Ground Floor]
3.) Main Build R5 - [Room no.-18, Ground Floor]

- Every Switches acts as access layer which helps to provide network connectivity to all classroom, lab, faculty room etc.
- Access layer directly Connect:
  ① Ethernet Port
  ② Wireless AP
  ③ Lab System
  ④ Outdoor Wireless AP (ex- 018)

→ Server Room Architecture / Topology



Topology :-

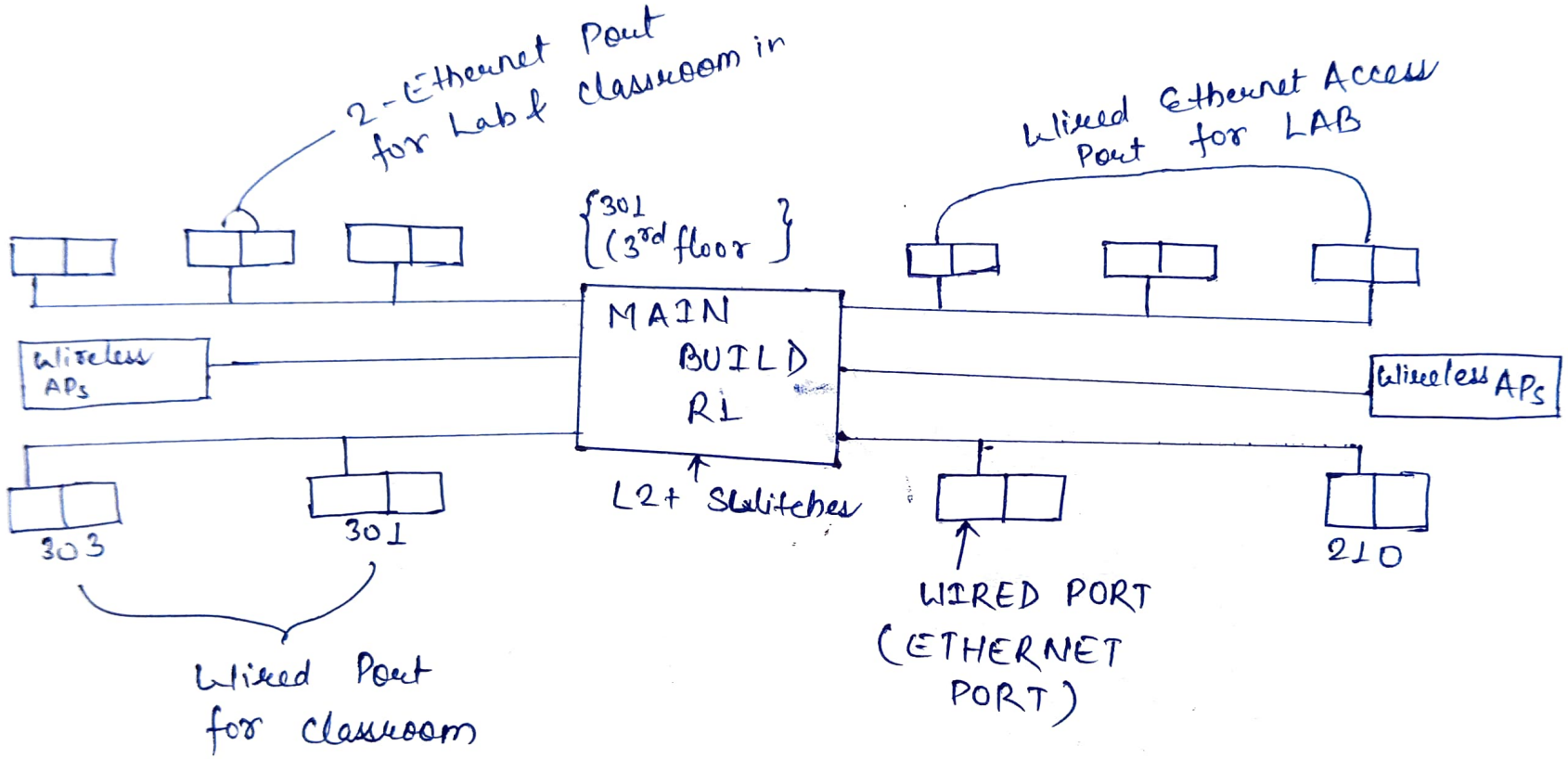


→ R1, R7, R2, R6 → Distribution layer

→ Server Room → Core Layer

→ R1, R2, R3, R4, R5, R6, R7 → Access Layer

Access layer Switch (3rd floor - 301)



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Windows PowerShell
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PS C:\Users\DELL> ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 11:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

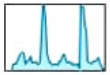
Wireless LAN adapter Local Area Connection* 12:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

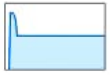
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:40e4:131b:66c7:c991:3c3:964c:aaba
   Temporary IPv6 Address. . . . . . : 2409:40e4:131b:66c7:95ff:830f:4fd6:4f0d
   Link-local IPv6 Address . . . . . : fe80::cdb6:245a:a163:49ca%10
   IPv4 Address. . . . . . . . . . . : 10.86.89.193
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::54f6:b6ff:fe1a:979c%10
                                       10.86.89.169

PS C:\Users\DELL>
```



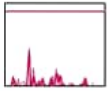
CPU
3% 1.60 GHz



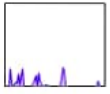
Memory
8.1/15.7 GB (52%)



Disk 0 (C: E: D: F:)
SSD (NVMe)
11%



Ethernet
Ethernet 2
S: 0 R: 0 Kbps



GPU 0
Intel(R) Iris(R) Xe Gra...
0%

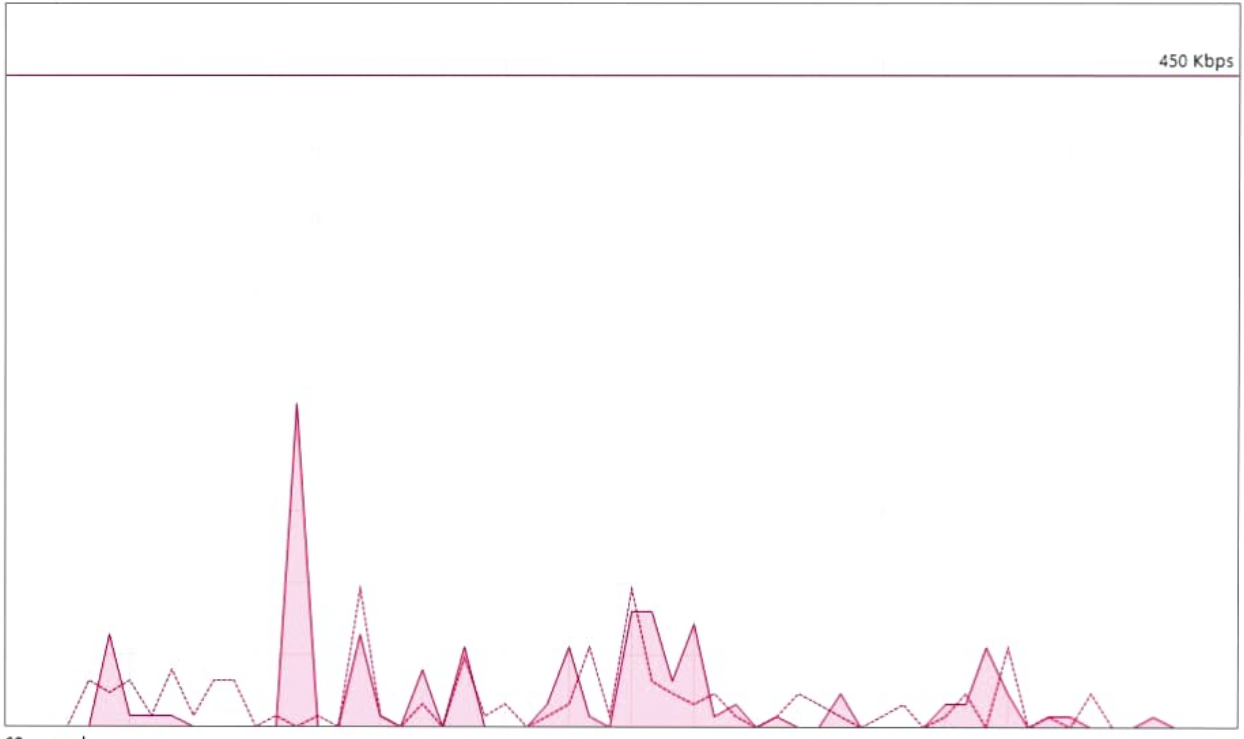
Ethernet

Remote NDIS based Internet Sharing Device

Throughput

500 Kbps

450 Kbps



60 seconds

0

Send
0 Kbps

Receive
0 Kbps

Adapter name:	Ethernet 2
Connection type:	Ethernet
IPv4 address:	10.60.18.214
IPv6 address:	2409:40e4:114b:de9b:e1c7:3588:b9d1:a121

```
Windows PowerShell
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PS C:\Users\DELL> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6fca:531a:400b:3e7b%41
   IPv4 Address. . . . . . . . . . . : 10.60.18.214
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.60.18.98

Wireless LAN adapter Local Area Connection* 11:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 12:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
PS C:\Users\DELL>
```

(6)

Difference b/w Wired & Wireless Speed ?

Wired Network	Wireless Network
• Faster Communication	• Slower Communication
• Stable Connection	• Affected by distance / Interference
• Low latency	• Higher latency
• Dedicated physical medium	• Shared radio medium.

- Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.
- Wireless connections uses radio waves which are affected by interference, distance obstacles & multiple connected users.

Q.) Role of Subnet mask in network ?

A Subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

- It is important for :
  1. Network Segment
  2. Efficient Routing
  3. Communication b/w devices.

OBSERVATION

During the study of the college network infrastructure :-

1. A centralized server room was used as network backbone.
2. Two L3 switches were connected in redundant cyclic structure.
3. Managed TP-Link POE+ switches were used in network cabinets.

Advantage of Existing Network?

- Scalable architecture.
- Centralized management.
- Supports POE+ device.
- Easy expansion of floors & classroom.
- Support both wireless & wired.
- Redundant link for failure.

CONCLUSION

The observed campus LAN follows a hybrid hierarchical topology consist of Tree, Extended Star, Ring. The network uses managed POE switches, wireless access point, Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.